

1. Square star pattern

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    for (int i=1; i<=5 ; i++) {
```

```
        for (int j=1; j<=5 ; j++) {
```

```
            cout << "*" ;
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns> cd "c:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 01_Square.cpp -o 01_Square } ; if ($?) { .\01_Square }
*****
*****
*****
*****
*****
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

2.

Hollow Square

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    for (int i=1; i<=5; i++) {
```

```
        for (int j=1; j<=5; j++) {
```

```
            if (i==1 || i==5 || j==1 || j==5)
```

```
                cout << "*" ;
```

```
            else
```

```
                cout << " " ;
```

```
        } cout << endl;
```

```
    } return 0;
```

```
> cd "c:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 02_Hollow_Square.cpp -o 02_Hollow_Square } ; if ($?) { .\02_Hollow_Square }
```

```
*****
```

```
*      *
```

```
*      *
```

```
*      *
```

```
*****
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

CHAT

SESSIONS



③ Triangle

#include <iostream>

using namespace std;

int main() {

for (int i=5; i>=1; i--) {

for (int j=1; j<=i; j++) {

cout << " * ";

}
cout << endl

}
return 0;

}

```
> cd "C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 03_Triangle.cpp -o 03_Triangle } ; if ($?) { .\03_Triangle }
*
**
***
****
*****
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

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SESSIONS



Inverted Triangle

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    for (int i = 5; i >= 1; i--) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << endl;
```

```
        }
```

```
    return 0;
```

```
}
```

```
> cd "c:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 04_Inverted_Triangle.cpp -o 04_Inverted_Triangle } ; if ($?) { .\04_Inverted_Triangle }
```

```
*****
```

```
*****
```

```
***
```

```
**
```

```
*
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

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SESSIONS



5. Pyramid

```
#include <iostream>
using namespace std;
int main() {
    for (int i=1; i<=5; i++) {
        for (int j=1; j<=5-i; j++)
            cout << " ";
        for (int k=1; k<=2*i-1; k++)
            cout << "* ";
        cout << endl;
    }
    return 0;
}
```

```
op\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++  
05_Pyramid.cpp -o 05_Pyramid } ; if ($?) { .\05_Pyramid }
```

```
*
```

```
***
```

```
*****
```

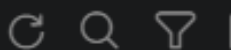
```
*****
```

```
*****
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\So  
urce_code\Star_Patterns>
```

✓ CHAT

SESSIONS



6. Diamond

```
#include <iostream>
using namespace std;
int main() {
```

```
    for (int i=1; i<=5; i++) {
        for (int j=1; j<=5-i; j++)
```

```
            cout << " ";
```

```
        for (int k=1; k<=2*i-1; k++)
```

```
            cout << "* ";
```

```
        cout << endl;
```

```
    }
```

```
    for (int i=4; i>=1; i--) {
```

```
        for (int j=1; j<=5-i; j++)
```

```
            cout << " ";
```

```
> cd "c:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 06_Diamond.cpp -o 06_Diamond } ; if ($?) { .\06_Diamond }
```

```
*
```

```
***
```

```
*****
```

```
*****
```

```
*****
```

```
*****
```

```
*****
```

```
***
```

```
*
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

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SESSIONS



7.

Inverted Half Diamond

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    for (int i=5; i>=1; i--) {  
        for (int j=1; j<=i; j++)
```

```
            cout << "*" << " ";
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
> cd "c:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 08_Half_Inverted_Diamond.cpp -o 08_Half_Inverted_Diamond } ; if ($?) { .\08_Half_Inverted_Diamond }
```

```
*****
```

```
*****
```

```
***
```

```
**
```

```
*
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

▼ CHAT

8. Half Diamond

```
#include <iostream>
using namespace std;

int main() {
    for (int i=1; i<=5; i++) {
        for (int j=1; j<=i; j++)
            cout << "*" << " ";
        cout << endl;
    }
    for (int i=4; i>=1; i--) {
        for (int j=1; j<=i; j++)
            cout << "*" << " ";
        cout << endl;
    }
    return 0;
}
```

① Number Poth

```
#include <iostream>
using namespace std;

int main() {
    // Number Poth
    // ...
}
```

```
> cd "c:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns\" ; if ($?) { g++ 07_Half_Diamond.cpp -o 07_Half_Diamond } ; if ($?) { .\07_Half_Diamond }
```

```
*
```

```
**
```

```
***
```

```
****
```

```
*****
```

```
****
```

```
***
```

```
**
```

```
*
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Star_Patterns>
```

✓ CHAT

SESSIONS



① Number Pattern 2 (Implementing)

#include <iostream>

using namespace std;

int main () {

for (int i=1; i<=5; i++) {

for (int j=1; j<=5; j++) {

cout << j << " ";

}

cout << endl;

}

return 0;

}

```
e\Source_code\Number_Patterns\" ; if ($?) { g++ 02_Incrementing.cpp  
-o 02_Incrementing } ; if ($?) { .\02_Incrementing }  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\So  
urce_code\Number_Patterns> 
```

② Number pattern 1 (square)

#include <iostream>

using namespace std;

int main() {

for (int i=1; i<=5; i++) {

for (int j=1; j<=5; j++) {

cout << "1";

3
cout << endl;

3
return 0;

3

```
ktop\3rd-Yr-Week-1-Module\Source_code\Number_Patterns\" ; if ($?) {  
g++ 01_Number_Square.cpp -o 01_Number_Square } ; if ($?) { .\01_Numb  
er_Square }
```

```
1 1 1 1 1
```

```
1 1 1 1 1
```

```
1 1 1 1 1
```

```
1 1 1 1 1
```

```
1 1 1 1 1
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\So  
urce_code\Number_Patterns>
```


③ Number Pattern 3 (Right Triangle)

```
#include <iostream>
using namespace std;
int main() {
    for (int i=1; i<=5; i++) {
        for (int j=1; j<=i; j++) {
            cout << " " << " ";
        }
        cout << endl;
    }
    return 0;
}
```

```
> cd "C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Number_Patterns\" ; if ($?) {  
g++ 03_Right_Triangle.cpp -o 03_Right_Triangle } ; if ($?) { .\03_Ri  
ght_Triangle }
```

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\So  
urce_code\Number_Patterns> █
```

Q1 Number Pattern 4 (Inverted Diamond)

#include <iostream>

using namespace std;

int main() {

int num = 3;

for (int i = 5; i >= 1; i--) {

for (int j = 1; j <= i; j++) {

cout << num << " ";

}

num++;

cout << endl;

}

return 0;

}

```
Ktop\3rd-Yr-Week-1-Module\Source_code\Number_Patterns\ , if ($?) {  
g++ 06_Incrementing_Diamond.cpp -o 06_Incrementing_Diamond } ; if ($  
?) { .\06_Incrementing_Diamond }
```

```
3  
4 4  
5 5 5  
6 6 6 6  
7 7 7 7 7  
6 6 6 6  
5 5 5  
4 4  
3
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\So  
urce_code\Number_Patterns>
```


Number Pattern 5 (Sandwich Pattern)

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    for (int i = 5; i >= 1; i--) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << (8 - i) << " ";
```

```
            cout << endl;
```

```
        }
```

```
        for (int i = 3; i <= 5; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                cout << (8 - i) << " ";
```

```
                cout << endl;
```

```
            }
```

```
        } return 0;
```

```
    }
```

```
Ktop\3rd-Yr-Week-1-Module\Source_code\Number_Patterns\ , if ($?) {  
g++ 05_Sandwich.cpp -o 05_Sandwich } ; if ($?) { .\05_Sandwich }  
3 3 3 3 3  
5 5 5  
7  
5 5 5  
3 3 3 3 3  
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\Source_code\Number_Patterns>
```

⑥ Number Pattern 6 (Incrementing Diamond)

#include <iostream>

using namespace std;

int main() {

for (int i=1; i<=5; i++) {

for (int j=1; j<=i; j++) {

cout << i+j << " ";

cout << endl;

}

for (int i=4; i>=1; i--) {

for (int j=1; j<=i; j++) {

cout << i+j << " ";

cout << endl;

}

return 0;

}

```
ktop\3rd-Yr-Week-1-Module\Source_code\Number_Patterns\" ; if ($?) {  
g++ 04_Inverted_Diamond.cpp -o 04_Inverted_Diamond } ; if ($?) { .\0  
4_Inverted_Diamond }
```

```
3 3 3 3 3
```

```
4 4 4 4
```

```
5 5 5
```

```
6 6
```

```
7
```

```
PS C:\Users\prajj\OneDrive\Documents\Desktop\3rd-Yr-Week-1-Module\So  
urce_code\Number_Patterns> 
```